Technology Stack (Architecture & Stack)

Date:	29 September 2026
Team ID:	SWTID-2026-6934
Project Name:	FitSense AI – Personalized Fitness Recommendations Powered by AI
Maximum Marks:	4 Marks

1. **TEAM ID:** SWTID-2026-6934
2. **TEAM LEADER:** SAMUEL JOSHUA R & samrajan0210@gmail.com
3. **TEAM MEMBERS:**
    - ROOHITH S & roohith0529@gmail.com
    - SAMVEL M & mssamvel777@gmail.com
    - SATHISHKUMAR B & baskersathishbaskersathish@gmail.com

Technical Architecture: ConnectHub follows a secure, scalable 3-tier MVC architecture. Client requests are handled by the Express.js gateway, authenticated via JWT middleware, processed through business logic controllers, and persisted using Mongoose models connected to a MongoDB database.

Table-1: Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	API Consumer interface for testing and interaction	Postman / React JS / Mobile App
2	Application Logic-1	User registration, password hashing, and JWT middleware	Node.js, Express.js
3	Application Logic-2	Contact CRUD operations, tag filtering, and search	Express Controllers & Mongoose Models
4	Database	NoSQL document storage for users and contacts	MongoDB
5	Cloud Database	Cloud database deployment option	MongoDB Atlas
6	File Storage	Environment configuration workspace	Local Filesystem / Environment Config (.env)

7	External API-1	Optional email notification service	Nodemailer / SendGrid API
8	Machine Learning Model	Not applicable for core contact utility	N/A
9	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud	Local Node Server / Cloud (Render/AWS)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Core open-source frameworks used	Node.js, Express.js, Mongoose
2	Security Implementations	Access controls and authentication security	bcrypt.js hashing, JWT authorization headers, CORS protection
3	Scalable Architecture	Architectural pattern for maintainability	3-tier MVC Architecture (Models, Routes, Controllers)
4	Availability	Server uptime and request handling	Stateless JWT sessions and environment port binding
5	Performance	Query optimization and fast retrieval	Indexed MongoDB fields for instant search and filtering